THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET B

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The value of $\cos 0^\circ + \sin 90^\circ + \tan 45^\circ$ is: 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 3

2. $(1 + \tan^2 A)\cos^2 A$ equals: 1

(a) 1 **(b)** $\sin^2 A$ **(c)** $\tan^2 A$ **(d)** 0

3. For $A = 30^\circ$, Student A says $\sin 2A = 2\sin A = 1$; Student B says $\sin 2A = \sin 60^\circ = \frac{\sqrt{3}}{2}$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

4. A 10 m tree casts a shadow $10\sqrt{3}$ m long. Student A says the sun's altitude is 30°; Student B says 60°. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

5. If the height of a tower equals the distance of the point of observation from its foot, the angle of elevation is: 1

(a) 30° **(b)** 60° **(c)** 45° **(d)** 90°

6. As the length of the shadow of a tower increases, the angle of elevation of the sun: 1

(a) remains the same **(b)** decreases **(c)** increases **(d)** becomes 90°

7. If tangents at P and Q meet at T and $\angle POQ = 140^\circ$, then $\angle PTQ$ is: 1

(a) 40° **(b)** 70° **(c)** 140° **(d)** 50°

8. A student states: "Two tangents can be drawn to a circle from a point inside it." The correct response is: 1

(a) Correct

(b) Only one can be drawn

(c) Infinitely many can be drawn

(d) Incorrect — no tangent can be drawn from an interior point

9. The length of the tangent from a point 17 cm from the centre of a circle of radius 8 cm is: 1

(a) 25 cm **(b)** 15 cm **(c)** 12 cm **(d)** 9 cm

10. A student states: "The area of a sector of angle θ is $\frac{\theta}{360^\circ} \times 2\pi r$." The correct response is: 1

(a) Correct

(b) Correct only for a semicircle

(c) Incorrect — that is the arc length; area $= \frac{\theta}{360^\circ}\pi r^2$

(d) Cannot be determined

11. The perimeter of a sector of radius r and angle θ is: 1

(a) $r + \frac{\theta}{360^\circ} \times \pi r^2$ **(b)** $2\pi r$ **(c)** $\frac{\theta}{360^\circ} \times 2\pi r$ **(d)** $2r + \frac{\theta}{360^\circ} \times 2\pi r$

12. If the circumference of a circle is 44 cm, the area of its quadrant is: 1

(a) 19.25 cm² **(b)** 154 cm² **(c)** 77 cm² **(d)** 38.5 cm²

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. If $\sin(A+B)=1$ and $\cos(A-B)=\frac{\sqrt{3}}{2}$, \$0^\circB, $find A and B$. **2**

14. A slide for children has its top at a height of 1.5 m and is inclined at 30° to the ground. Find the length of the slide. **2**

15. Prove that the tangents drawn at the ends of a diameter of a circle are parallel. **2**

16. Find the area of a sector of a circle of radius 4 cm with central angle 30°. (Use $\pi = 3.14$) **2**

17. Find the angle subtended at the centre of a circle of radius 7 cm by an arc of length 11 cm. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. In $\triangle ABC$, right-angled at B, $AB = 24$ cm and $BC = 7$ cm. Determine **3**
(i) $\sin A, \cos A$
(ii) $\sin C, \cos C$.

19. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked. **3**

20. Prove that the chord of the larger of two concentric circles which touches the smaller circle is bisected at the point of contact. **3**

21. A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the corresponding minor segment. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) **3**

22. A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top is 60°. From another point 20 m away from this point on the line joining it to the foot of the tower, the angle of elevation is 30°. Find the height of the tower and the width of the canal. **3**

OR

The angles of depression of the top and the bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45°. Find the height of the multi-storeyed building and the distance between the two buildings. **3**

23. The perimeter of a sector of a circle of radius 5.2 cm is 16.4 cm. Find the area of the sector. **3**

OR

A circular park of radius 28 m has a sector-shaped flower bed with angle 45°. Find the area of the bed and the cost of turfing it at ₹4 per m². **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** In $\triangle ABC$, right-angled at B, $AB = 12$ cm and $BC = 5$ cm. **5**
(i) Find AC.
(ii) Write all six trigonometric ratios of $\angle A$.
(iii) Verify that $1 + \tan^2 A = \sec^2 A$.

OR

If $\sin\theta + \cos\theta = p$ and $\sec\theta + \csc\theta = q$, show that $q(p^2 - 1) = 2p$. **5**

25. A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are 8 cm and 6 cm. Find the sides AB and AC. **5**

OR

Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. Hence find the length of the tangent from a point 15 cm from the centre of a circle of radius 9 cm. **5**

END OF QUESTION PAPER — **Total: 50 Marks**